Venkata Lakshmi Parimala Pasupuleti

Software Engineer · AI-Powered Test Automation · ML Infrastructure · Generative AI · CI/CD · Scalable Test Systems · Observability · Python · LangGraph · FastAPI

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PROFESSIONAL SUMMARY

Software Engineer with 8 years of production Python experience and two shipped Generative AI systems - including AgentProbe, an open-source AI-powered test infrastructure toolkit built to solve the same problem this role targets: enabling faster release cycles and higher quality software through automated evaluation. AgentProbe's CI-integrated regression gate, 5-metric quality assessment engine, and LLM-as-judge auto-rater enforce software quality at every merge - the same pattern Google Platforms & Devices is building at device scale. At Entrata, owned large-scale data processing pipelines (180,000 domain records per release cycle), optimized 200+ SQL queries for scalable performance (30% gain), and built ML infrastructure for debugging and observability that reduced triage time 40% across 100K+ monthly active users. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Engineering & Backend: Python, SQL, FastAPI, PHP, JavaScript, TypeScript, Bash, Docker, Microservices, Scalable Systems, AWS (S3, EC2, RDS, Glue), Redis, RabbitMQ, REST APIs, CI/CD (GitHub Actions), Test Infrastructure, Release Automation, PostgreSQL, MySQL, Oracle, Linux, Git

Generative AI & ML: Generative AI, LLM Orchestration, LangGraph, LangChain, Agentic AI, ML Infrastructure, Model Optimization, Model Evaluation, RAG Pipelines, ChromaDB, LLM-as-Judge, Hallucination Detection, Agent Observability, HITL, Prompt Engineering, OpenAI API, Anthropic API, MCP

ML & Analytics: Data Processing, PyTorch, scikit-learn, NLP, BERT, spaCy, PySpark, Feature Engineering, Quality Metrics, Statistical Modeling, Tableau, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an agentic Generative AI mission-planning system (LangGraph stateful graph): parse → plan → retrieve → validate → conditional routing - automated quality checks enforcing FAA altitude limits, battery budgets, and geometric no-fly-zone constraints before export.
- Implemented RAG pipeline with model optimization for latency and accuracy; validation guardrails with cited-source output grounding; automated test infrastructure for compliance verification across mission plans.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets - $R^2 = 0.977$ regression accuracy; full data processing pipeline from ingestion through feature engineering, model optimization, and statistical validation.
- Delivered Python and SQL workshops to 600+ students; designed scalable Tableau dashboards improving research quality and comprehension by 40% across two cohorts.

Entrata - PropTech / Affordable Housing SaaS
Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Built ML infrastructure and observability tooling** that reduced debugging time 40% and kept software quality below 10 bugs for 6+ consecutive months - internal test automation eliminating manual verification across compliance-critical workflows.
- **Owned large-scale data processing pipelines end-to-end** - fragmented federal source data expanded into ~180,000 scalable domain records; validation and reconciliation at every stage; zero compliance violations across 7 years of ownership at 99.5% uptime.
- **SQL model optimization at scale** - optimized 200+ queries (30% performance gain) through execution plan analysis, indexing, partitioning, and stored procedure refactoring across multi-tenant production schemas.
- **Shipped 20+ full-stack production modules** (Python, PHP, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints; led Docker microservices redesign - 50% scalability improvement, 30% release cycle reduction; eliminated 80% of manual operational workload through automation.
- Mentored 5+ junior engineers; 300+ peer code reviews; AWS S3 migration with 100% PHI compliance (zero security incidents).

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built property management modules (PHP, Python, PostgreSQL); data processing pipeline for 1.6M HUD/Tax Credit records; AWS S3 cloud migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: AI-Powered Test Infrastructure & Quality Automation Test Infra · CI/CD

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Open-source AI-powered test infrastructure built to enable faster release cycles and higher quality software: 15-endpoint FastAPI backend, 5-metric quality assessment engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate blocking quality regressions before merge, LLM-as-judge auto-rater, pass@k harness, and multi-provider LLM routing for cost optimization. Used to evaluate LeaseI across 1,240+ compliance test cases - zero regressions shipped to production.

LeaseI: 8-Agent Generative AI Investigation Platform GenAI · Agentic

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Generative AI agentic system with 8 specialized agents and conditional routing - LLM inference over fragmented data sources producing ranked, evidence-backed hypotheses with HITL escalation when model confidence is below threshold. 93.6% precision, 87.9% F1 across 1,240 test cases; CI-integrated quality gate with LLM-as-judge scoring.

AI on FHIR: NLP-Driven Medical Records Query System NLP · LLM

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask + React NLP platform using LLM inference and BERT embeddings to translate natural-language queries into FHIR API calls - reduced manual retrieval time ~60%; scalable data processing across structured medical records with HIPAA-compliant RBAC.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017